Thermodynamic description of the Cu–Mg system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Cu–Mg binary system as calculated with pycalphad from the SGTE dataset CuMg-16Hal.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → (Cu) + Cu2Mg, lies at 725.2 °C.

1. Introduction

The Cu–Mg system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as congruent compound, intermetallic rich. The thermodynamic description is the SGTE collection entry [1], which reproduces the assessment of B. Hallstedt, unpublished, 2016. [2]. The calculations use pycalphad 0.11.2 at 1 atm.

Pure Cu melts at 1084.6 °C. Pure Mg melts at 649.9 °C.

2. Phase diagram and invariant reactions

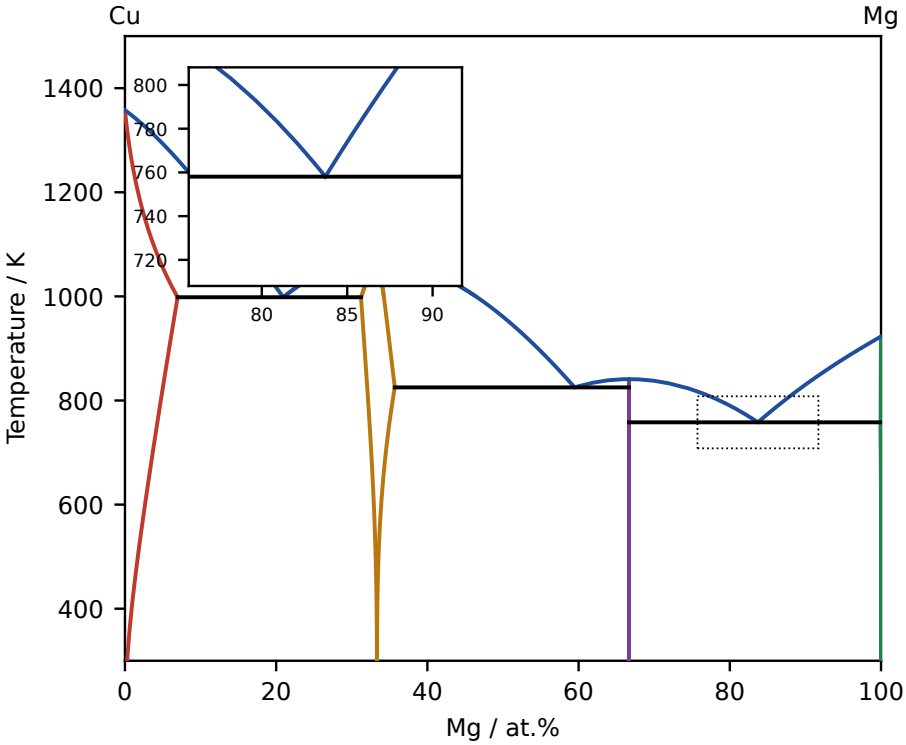


Fig. 1. Calculated Cu–Mg phase diagram (inset: enlarged view near the lowest invariant).

Eutectic equilibrium L → (Cu) + Cu2Mg is found at 725.2 °C and 20.9 at.% Mg (L). We obtain 552.0 °C for the eutectic reaction (L → Cu2Mg + CuMg2), where L has 59.5 at.% Mg. Eutectic equilibrium L → CuMg2 + (Mg) is found at 484.9 °C and 83.7 at.% (L). Cu2Mg melts congruently at 795.8 °C and 33.0 at.% Mg. CuMg2 melts congruently at 567.9 °C and 66.5 at.%

Mg.

Table 1 compares the calculated temperatures with experimental values from the literature [3–5]; in this synthetic document those values and references are fictitious.

Table 1. Invariant reactions and congruent points, calculated and experimental (Cu–Mg).

Reaction	Type	T (K) this work	T exp.	at.% Mg, 1	at.% Mg, 2	at.% Mg, 3
L → Cu2Mg	congruent melting	1068.9	1064.9 [5]	33.0	33.0	–
L → (Cu) + Cu2Mg	eutectic	998.4	–	20.9	7.0	31.2
L → CuMg2	congruent melting	841.1	843.1 [4]	66.5	66.5	–
L → Cu2Mg + CuMg2	eutectic	825.2	834.2 [5]	59.5	35.7	66.7
L → CuMg2 + (Mg)	eutectic	758.1	761.1 [3]	83.7	66.7	100.0

Experimental temperatures are fictitious values attributed to invented references [3–5] for testing evidence-class extraction; they are not data.

3. Phase stability

Table 2. Solid phases: structure, stability range and homogeneity range (Cu–Mg).

Phase	Pearson	SB	Prototype	T min (°C)	T max (°C)	Range (x(Mg))	T (°C)
(Cu)	cF4	A1	Cu	< 27	1084.6	0.000–0.069	727
Cu2Mg	cF24	C15	MgCu2	< 27	795.8	0.320–0.356	557
CuMg2	oF48	Cb	Mg2Cu	< 27	567.9	0.667	47
(Mg)	hP2	A3	Mg	< 27	649.9	1.000	487

The upper stability limit of Cu2Mg is 795.8 °C. The solubility of Mg in (Cu) reaches 7.0 at.% at 725.2 °C.

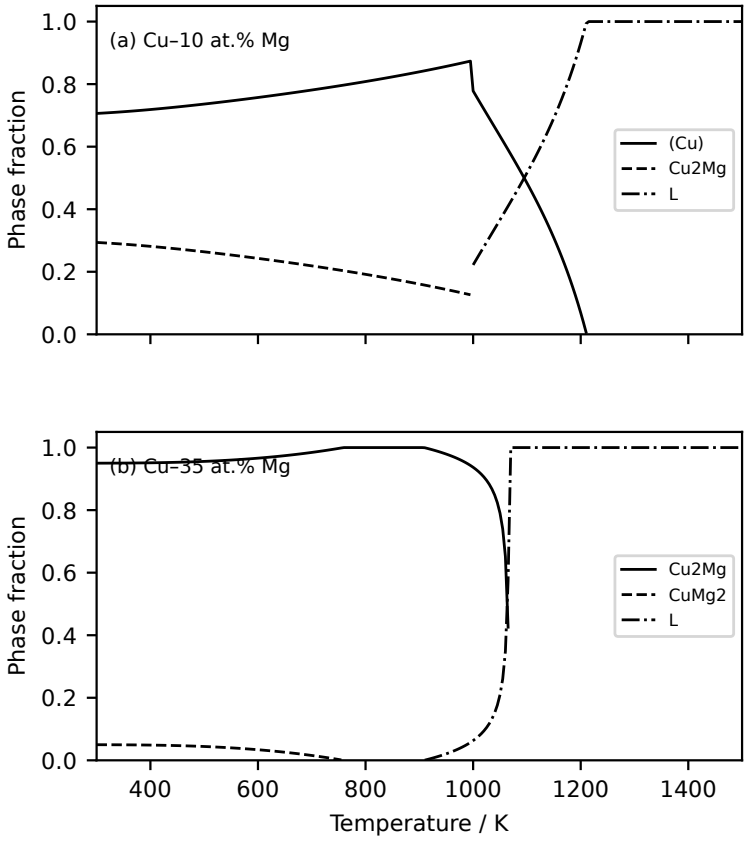


Fig. 2. Calculated equilibrium phase fractions versus temperature for Cu–10 at.% Mg and 35 at.% Mg.

4. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −9350 J/mol, at 44.0 at.% Mg. The calculated enthalpy of formation of CuMg2 at 298.15 K is −8.65 kJ/mol per mole of atoms. Per formula unit CuMg2 this is −25.9 kJ/mol. Its heat capacity at room temperature is 24.59 J/(mol K). For the equiatomic liquid at 1450 K, the Gibbs energy of mixing is −15.88 kJ/mol-atom.

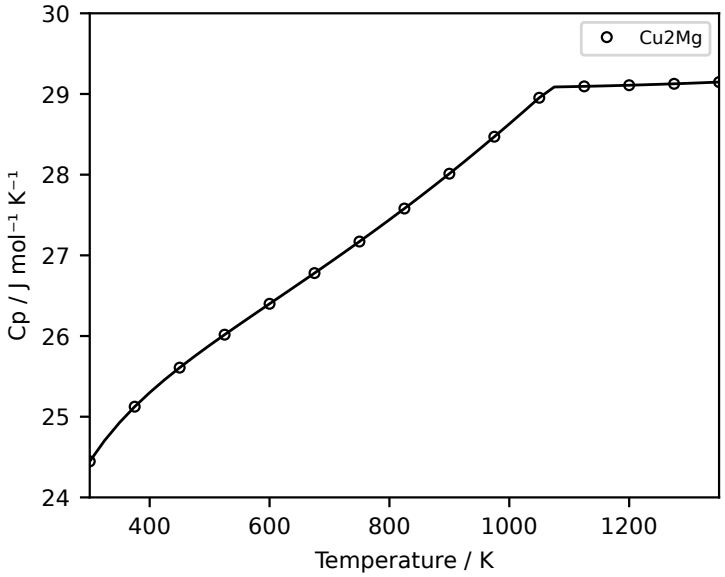


Fig. 3. Calculated heat capacity of Cu2Mg.

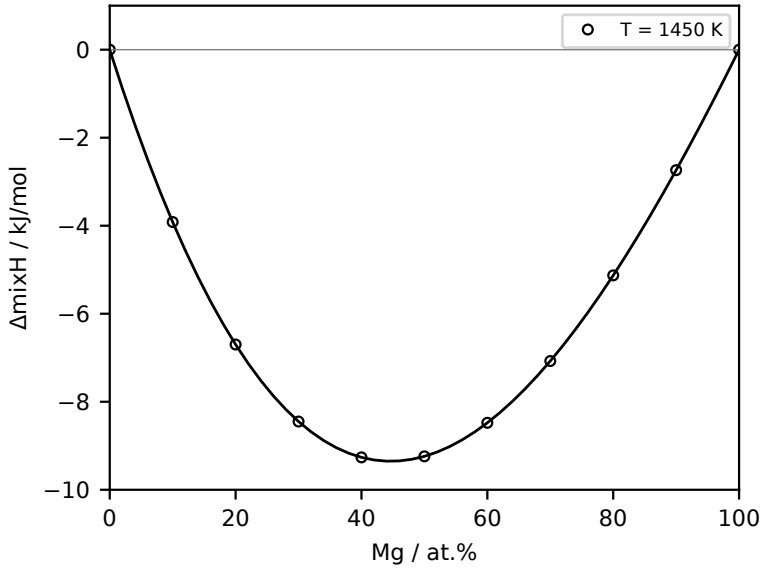


Fig. 4. Calculated mixing enthalpy of the liquid at 1450 K.

Table 3. Calculated formation properties of the intermediate phases at 298.15 K (Cu–Mg).

		298.15 K, per mole of formula units		
Phase	**at.% Mg**	**ΔfH / kJ mol⁻¹**	**ΔfS / J mol⁻¹ K⁻¹**	**Cp / J mol⁻¹ K⁻¹**
Cu2Mg	33.5	−30.59 ± 0.5	5.89	73.28
CuMg2	66.7	−25.95 ± 0.5	2.01	73.78

Reference: stable elements (SER) at 298.15 K; values per mole of formula units (formula unit as written in the Phase column). ± values are synthetic illustration uncertainties, not data.

Table 4. Integral and partial mixing properties of the liquid at 1450 K (Cu–Mg).

x(Mg)	ΔmixH / kJ mol⁻¹	ΔmixG / kJ mol⁻¹	a(Cu)	a(Mg)
0.100	−3.92	−7.22	0.8625	0.0095
0.300	−8.45	−14.37	0.5010	0.0944
0.500	−9.24	−15.88	0.2261	0.3175
0.700	−7.07	−12.99	0.0826	0.6242
0.900	−2.74	−6.04	0.0184	0.8934

5. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.

References

[1] B. Hallstedt, SGTE binary dataset collection, Calphad 89 (2025) 102833 (CC BY 4.0).

[2] B. Hallstedt, unpublished, 2016.

[3] K. Synthetica, L. Exemplar, J. Fict. Phase Equilib. 12 (1987) 345–352 — fictitious reference, invented for this synthetic benchmark.

[4] M. Placeholder, Z. Notreal, Fict. Thermochim. Acta 41 (1994) 77–89 — fictitious reference, invented for this synthetic benchmark.

[5] R. Imaginary, Proc. Synth. Calorim. Soc. 8 (2003) 1102–1110 — fictitious reference, invented for this synthetic benchmark.